

Edge-Computing Video Analytics for Real-Time Traffic Monitoring

Critical Analysis of the Liverpool Smart Pedestrians Project

A. Introduction and Objectives

The city of Liverpool, New South Wales, wanted to tackle the urban challenges of increasing pedestrian density, congestion, and the need for data-driven mobility planning while still complying with privacy requirements. This is where the Liverpool Smart Pedestrian project comes in, a real-time traffic monitoring system using edge-computing video analytics. The main idea was to help urban planners support the rapidly growing city of Liverpool by collecting accurate data on pedestrian, vehicle, and cyclist movements throughout the city. Rather than installing new infrastructure, the project relied on existing CCTV cameras to gather reliable movement data. This allowed city planners to respond to the growing population without breaching privacy regulations. By converting traditional CCTV systems into intelligent sensors, the project aimed to collect useful data while keeping the costs and data privacy risks low.

B. Methodology

The project's goal was essentially to find and track different types of transportation in real time by building an edge-computing visual sensor. The system's design was influenced by input from community workshops and discussions with urban planners, especially in privacy protection, system scalability, and compatibility with existing infrastructure. This worked as the sensor processed video streams locally, allowing only anonymous metadata, like object counts and movement paths, to be tracked. This minimized privacy concerns and reduced data transmission requirements. This also accounted for practical limitations such as limited bandwidth when using

LoRaWAN, the need for real-time processing, and the challenge of deploying hardware in outdoor environments. The city of Liverpool, New South Wales wanted to tackle the urban challenge of an increasing pedestrian density, congestion, and the need for data-driven mobility planning while still following privacy requirements. This is where the Liverpool Smart Pedestrian project comes in, a real-time traffic monitoring system using edge-computing video analytics. The main idea was to help urban planners to support the rapidly growing city of Liverpool, by collecting accurate pedestrian, vehicle, and cyclist movement throughout the city. Rather than installing new infrastructure, the project relied on existing CCTV cameras to gather reliable movement data. This allowed for city planners to respond to the growing population without breaching privacy regulations. By converting traditional CCTV systems into intelligent sensors, the project aimed to collect useful data while keeping the costs and data privacy risks low.

C. Technology and Implementation

The system basically used an NVIDIA Jetson TX2 for its main hardware as it is powerful, saves energy, and can run AI models directly on the device itself. In order to detect people, bicycles, and vehicles, the YOLO V3 was chosen as it works quickly and is also accurate, both being important factors for real-time use. In addition, in order to track moving objects, the SORT tracking algorithm was used, which is actually able to track multiple objects at the same time. Perhaps most notably, all of the video processing happened on the device itself instead of sending it to the cloud, thanks to edge computing. This in turn decreased network usage and also maintained privacy.

D. Validation and Performance

The Oxford Town Center was used to test the system, which found the average accuracy to be 69% and the processing speed to be around 20 frames per second, which is fast enough for monitoring in real-time. One issue was that sometimes fewer people were counted in crowded scenes because people were blocking each other in the video. Otherwise, in terms of system performance, there turned out to be stable CPU, GPU, memory usage and temperature during the real-world scenario. This proved that the system could successfully run continuously.

E. Real-World Applications

Two real-world scenarios were used to test the system. Firstly, an indoor test was conducted to see how well the system could detect unusual traffic and movement during an evacuation. The system performed well on this test. Second, an outdoor test was performed to show that the system could accurately track pedestrian, bicycle, and vehicle activity over a number of days. These tests show how the system can support urban planning by helping to ensure pedestrian safety, efficient infrastructure planning, and by improving overall traffic analysis.

F. Challenges and Future Work

Some of the main challenges were lower accuracy in crowded areas, slower tracking because of the inherent CPU lack of capacity, and unused GPU capacity. The researchers made some suggestions such as using more powerful hardware such as the NVIDIA Xavier and optimizing tracking algorithms to be able to run on the GPU. In addition, one note-worthy point is that this project was conducted in 2019. Many more new technologies have been developed since then such as newer YOLO versions, edge AI hardware, 5G networks, etc.

G. Personal Evaluation

Overall, the project showed how edge-computing video analytics can use CCTV systems to greatly improve urban planning by focusing on privacy, real-world testing, and the coordination of multiple systems.

References

Barthélemy, J., Verstaeevel, N., Forehead, H., & Perez, P. (2019). *Edge-computing video analytics for real-time traffic monitoring in a smart city*. *Sensors*, 19(9), 2048.

<https://doi.org/10.3390/s19092048>